

PHONOLOGICAL ENCODING AND THE PHONOLOGY OF SPANISH — THE ROLE OF THE SYLLABLE

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Abstract

During speech comprehension, listeners engage in a process known as phonological encoding. Phonological encoding comprises processes such as assigning gradient acoustic information to discrete phonemic categories (**categorization**), detecting possible word onsets in a continuous signal (**segmentation**), and locating words in the lexicon (lexical activation, **competition**, and **recognition**). According to some, evidence gathered from tasks that tap into phonological encoding provide crucial information for our understanding of speakers' mental representations (i.e., phonology). In addition to investigating speech production through articulatory and acoustic studies, laboratory phonologists (and psycholinguists) conduct experimental research on phonological encoding. This chapter reviews the experimental literature on one important phonological structure of Spanish, the syllable. The encoding evidence suggests that listeners whose native language is Spanish deploy their knowledge of their language's syllabification patterns in order to segment the speech chain into processable chunks, and to activate (and deactivate) word entries in the mental lexicon. This suggests that Spanish speakers' knowledge of the phonology of their language includes the syllable. Most aspects of Spanish speakers' phonological knowledge, such as assimilation or lenition rules, remain to be investigated from the perspective of phonological encoding, even though laboratory tasks are available and comparable evidence from other languages exists.

INTRODUCTION

Speech is fast, continuous, variable, and nonunique; yet, vocabularies are comprised of discrete entries, words. Spoken language is not divided into chunks separated by pauses—speech does not typically cue word boundaries—and listeners must learn to cope with this by developing segmentation strategies (i.e., they must deal with the “segmentation” problem). The representation of words in the lexicon is (perhaps) abstract and discrete, but two iterations of the same word pronounced by two different speakers are never acoustically identical—listeners must therefore learn to cope with extreme phonetic variability (i.e., the “lack of invariance” problem). Words often have other words embedded in them (e.g., *star* is embedded in *start*) or are phonologically similar to each other (e.g., *admiral* and *admiration* differ only after their fifth segment, ignoring stress), which further demonstrates that the fact that listeners typically recognize the words intended by speakers is nothing short of miraculous (i.e., the “nonuniqueness” problem)—see Cutler (2012), especially chapter 2, for an overview. The evidence to date demonstrates that spoken word recognition “is an active process of hypothesis construction. Listeners do not simply wait and understand utterances once they end. Nor do they even wait until the end of a word. Multiple word candidates, potentially consistent with the incoming speech, are considered concurrently” (Cutler, 2012, p. 40). In order to segment the speech signal into processable chunks and to access the phonological forms of words in the lexicon (and discard competitors), listeners utilize as many cues as they can. Most of these cues are gradient, probabilistic. It follows that the strategies listeners use must be expressed in probabilistic terms. Laboratory phonologists are interested in discovering what such strategies are, since understanding their nature is potentially akin to understanding (one aspect of) what listeners know (or can find out) about the sounds of their language, a fundamental question in the field of phonology. (Psycholinguists are interested in them for different reasons.) It turns out that many such strategies are language-specific because they depend on the composition of a language’s vocabulary. This suggests that much of this knowledge is learned rather than innate.

While listeners care about meaning, not sounds, it is sounds that they are met with, and it is only through the successful encoding of sounds into abstract representations that they can access meaning. The constellation of strategies listeners use to deal with sounds in real time—i.e., to encode them into abstract representations and, ultimately, words—is hereafter referred to as *phonological encoding*. The present chapter is motivated by the informed assumption that phonological encoding is informative to those who want to understand what speakers know about the sounds of their language(s). Evidence from the world of phonological encoding can contribute its own set of facts to whatever set of facts formal analyses can provide us with in favor of (or against) traditional phonological structures. Understanding phonological structure entails gathering as many facts as possible from as many paradigms as possible. Excluding *a priori* an entire set of facts or dismissing them as “performance phenomena” is akin to arbitrarily limiting our knowledge.

This chapter is about a single aspect of the phonological knowledge of Spanish speakers—the syllable. The syllable organizes segments into a hierarchical structure, and there is ample formal evidence that syllables are part of the phonological knowledge of Spanish speakers (Colina, 2009). Many phonological processes, for instance, are best expressed as operations over segments located in specific nodes on a hierarchical structure. But, is there evidence from phonological encoding—i.e., from the world of speech comprehension—that syllables are part of

the phonological knowledge of Spanish speakers? Does syllabic knowledge modulate how listeners who speak Spanish segment a continuous signal into chunks or how they access their lexicon (i.e., activate and deactivate word candidates)? The answer appears to be (for the most part) yes. Rather than provide an overview of what aspects of Spanish phonology have been studied from the perspective of phonological encoding, this chapter focuses on a single aspect of Spanish phonology. Nevertheless, various laboratory techniques are illustrated. Other aspects of the phonology of Spanish have received some attention from this perspective, such as lexical stress (Soto-Faraco, Sebastián-Gallés, & Cutler, 2001), but most have not. In fact, in order to illustrate the facts about the syllable, this literature review must go beyond Spanish and include studies on Catalan, French, Italian, and English. This should provide relevant context. Perhaps identifying this void will motivate future laboratory research on the phonology of Spanish.

The remainder of this chapter assumes that, in the process of comprehending speech in real time, segmentation strategies and lexical activation processes are different sets of processes—although obviously not independent from each other, the strategies used to break up the signal into chunks and those used to activate lexical candidates are in principle of a different nature (Norris & Cutler, 1985). The chapter is organized into two main sections, one focusing on segmentation-based evidence and one focusing on activation-based evidence. This division is pragmatic for the purposes of presentation and discussion. The relation between these two sets of processes needs to be better understood. In terms of the role of syllabification in each set of processes, four alternatives present themselves: on the one hand, it is possible that syllabic information is used during speech segmentation but not lexical activation, or it may be used in lexical activation but not segmentation; on the other hand, it could be exploited by both sets of processes, or it could not be used at all.

Syllabic segmentation: word-spotting and syllable monitoring

This section reviews some of the extant evidence supporting the hypothesis that listeners deploy their syllabic knowledge during speech segmentation—i.e., syllabification as a segmentation strategy. It will be seen that there is reason to believe that this is so, but not in all languages. In particular, the evidence suggests that French speakers employ syllable-based segmentation strategies, but English speakers do not. The evidence regarding Romance languages other than French, including Spanish, is more nuanced. The experimental studies reviewed in this section employ two types of comprehension-oriented psycholinguistic tasks: (i) the word-spotting task of Cutler & Norris (1988), and (ii) a version of the syllable-monitoring task (Mehler, Dommergues, Frauenfelder, & Seguí, 1981).

Let us begin with the word-spotting evidence. The only word-spotting study exploring the role of the syllable during speech segmentation was run on French (Dumay, Frauenfelder, & Content, 2002). At present, there is no comparable data for Spanish, but the syllabic patterns of French and Spanish are similar, and the French results should be extrapolatable to Spanish, at least in principle. Dumay et al. (2002) hypothesize “that syllable onsets help the listener segment the speech flow” (page 147) by providing clues to word onsets. Since word onsets necessarily coincide with syllable onsets, knowledge about possible (and impossible) syllable onsets should aid the task of finding possible word onsets or extracting possible words from a continuous signal. For instance, in French (like in Spanish), a word may begin in /bl/ but not /nl/; this, in turn, means that syllables in French may begin with /bl/ but not /nl/. By hypothesis, when French

speakers hear an /nl/ sequence, they assign the /n/ to the coda and the /l/ to the onset of separate syllables; but when they hear a /bl/ sequence they may assign both consonants to the same syllable as an onset cluster. Dumay et al.'s (2002) question was as follows: Does syllable-based phonotactic information aid French listeners during online speech segmentation? If so, this would provide evidence that French listeners have some type of syllabic knowledge and that they employ this knowledge during speech segmentation.

The researchers played disyllabic nonword sequences with intervocalic consonant clusters to a group of French-speaking listeners. Real French words, such as *lac* 'lake,' were embedded in the second half of nonwords, and the composition of the intervocalic consonant clusters was manipulated. For instance, listeners heard nonwords such as [zunlak] and [zublak]. If French listeners exploit syllable-based phonotactic information, the argument goes that it should be easier for them to spot the word *lac* 'lake' in [zunlak] than in [zublak]. Indeed, it was found that *lac* was more likely to go undetected in [zublak] than in [zunlak] nonwords, and that, when correctly spotted, detection was much faster (± 100 milliseconds) in [zunlak] than in [zublak]. It appears that, when hearing a nonword sequence such as [zublak], French listeners have the tendency to automatically syllabify it as [zu.blak], which hinders their identification of *lac* 'lake' by creating a situation in which word onset and syllable onset are not aligned.

The evidence that listeners use syllable-based segmentation strategies comes not only from the word-spotting experiment discussed above, but also from multiple syllable-monitoring experiments on French, starting with Mehler, Dommergues, Frauenfelder, & Seguí (1981). Comparable syllable-monitoring experiments have been conducted on Catalan, Italian, and Spanish. The articles will be discussed in approximate chronological order, and it will be seen that, whereas the French (and, perhaps, the Catalan) data may have a straightforward interpretation, the Spanish (and the Italian) data do not.

Mehler et al. (1981) designed a monitoring experiment in which French-speaking listeners were asked to press a button as soon as they identified, from a list of words, a sequence of sounds comprised of either two or three segments. Two-segment sequences were composed of a consonant and a vowel—e.g., participants were asked to monitor for sequences such as /ba/ or /pa/. Similarly, three-segment sequences had a consonant, a vowel, and a consonant—e.g., participants were asked to monitor for sequences such as /pas/ or /bal/. By hypothesis, participants would create a syllabic “mental image” of the sequence they were instructed to monitor for, which means that, without having been explicitly instructed to do so, participants would be monitoring for syllables rather than segment sequences (at least at some level). The target words played to the participants could have one of two key forms: they could begin in an open (or vowel-final) or in a closed (or consonant-final) syllable. This included, for instance, words such as *ballon* 'ball' and *balcon* 'balcony', or *palace* 'palace' and *palmier* 'palm tree.' Target words were presented in isolation rather than embedded in carrier phrases.

The first finding was that the participants could detect /ba/ in both *balcon* and *ballon*, and they could detect /bal/ in both *ballon* and *balcon* as well—i.e., they pushed the recognition button in all cases. This shows that, as expected, they understood the task to be one in which they were to monitor for segment sequences, not full syllables. The most important finding lies in the time it took for them to react to the words as a function of the sequence they were asked to monitor for. First, listeners detected /ba/ faster in *ballon* than in *balcon*. Second, they detected /bal/ faster in *balcon* than in *ballon*. This was called the “target-type by word-type interaction” (hereafter, target-by-word interaction), and finding such an interaction was key. In a second

experiment, listeners monitored for vowel + consonant sequences, such as /al/, in words that consistently began with a consonant, such as *ballon* and *balcon*. It was found that participants were faster at detecting /al/ in *balcon* than in *ballon*. It seems that it was easier for these listeners to recognize a sequence of two segments when they are part of the same syllable (i.e., the rhyme) than when they are not. In French, the /l/ in *balcon* is syllabified as the coda of the first syllable while the /l/ in *ballon* is syllabified as the onset of the second syllable.

The findings are in line with the following interpretation: As the auditorily presented words are being played to the listeners, they are being automatically encoded into sublexical phonological units. One of such sublexical units is, arguably, the syllable. A word such as *balcon* is divided into two syllables, and the /l/ is assigned to the first syllable because /lk/ is not a possible onset. If participants are monitoring for /ba/, the mismatch between what they are monitoring for and what they automatically encode from the signal (*bal.con*) triggers a recognition delay. The same is true if they are monitoring for /bal/ and they hear *ballon* (*ba.llon*)—i.e., the mismatch triggers a delay. This finding clearly suggests that French-speaking listeners encode auditorily presented words in chunks that look very much like syllables. These findings led Mehler and colleagues to propose that, since all languages have syllables, online syllabic encoding is a fundamental, universal aspect of speech comprehension. While the findings for French are still standing (but see Content, Meunier, Kearns, & Frauenfelder, 2001), the cross-linguistic scope of the “syllabic hypothesis” was short-lived. In any case, as Cutler, Mehler, Norris, & Seguí (1986) put it, “for our present purposes what is important in these studies is that listeners are clearly capable of extracting syllabic units from continuous speech when required to do so” (page 386). This suggests that syllabification is, at one level of representation, part of the phonological knowledge of French speakers.

Syllabification patterns differ across languages, and they are more complex and variable in some languages than they are in others. This may lead to cross-linguistic differences in syllabic knowledge, unless syllabic segmentation is a true universal. Syllabification patterns in French are relatively simple and consistent, and French listeners seem to have relatively robust syllabification intuitions (Content, Kearns, & Frauenfelder, 2001). In English, syllabification patterns are less transparent than in French, and English speakers are less consistent when explicitly asked to syllabify English words. One syllabic difference between French (and Spanish) and English is ambisyllabicity. In English, an intervocalic consonant preceding an unstressed vowel is potentially ambisyllabic—e.g., the /l/ in *palace* might be part of both the first ([pæl]) and the second ([ləs]) syllables, since both [pæl] and [ləs] (but neither [pæ] nor [əs]) are well-formed English syllabic units. Cutler et al. (1986) asked whether the syllable-monitoring evidence presented in Mehler et al. (1981) using French materials would prove reliable in a language such as English, with widespread ambisyllabicity. They replicated Mehler et al.’s (1981) task with English materials and English-speaking listeners.

Cutler et al. (1986) reported on four experiments. The first experiment focused exclusively on English. English-speaking listeners were asked to monitor for sequences comprising two or three segments, such as /ba/ and /bal/. The word list played to the listeners contained words such as *balance* and *balcony*. The /l/ in *balcony* is syllabified with the first syllable as its coda, but the /l/ in *balance* is potentially ambisyllabic. There was no clear prediction for words such as *balance*, since predictions came from the French results in Mehler et al. (1981) and ambisyllabicity is not a property of French phonology. But, if English listeners use syllabification during speech segmentation, their behavior should align with that of French

listeners in words such as *balcony*—they should detect /bal/ faster than /ba/ in such words. This experiment, however, led to a null result—in other words, there was no target-by-word interaction. English listeners did not detect /bal/ faster in *balcony* than in *balance*; and, what is more telling, they did not detect /ba/ faster in *balance* than in *balcony*.

The difference between the English (Cutler et al., 1986) and the French (Mehler, Dommergues, et al., 1981) results lend themselves to various interpretations. Perhaps French listeners use syllabification during speech segmentation and English listeners do not, which would mean that syllabic segmentation is not universal. Or perhaps the French materials—that is, the auditory stimuli used in Mehler et al. (1981)—contain important subsegmental information that facilitate the exploitation of syllabic segmentation strategies and the English materials do not. In order to test whether the French auditory stimuli were special in any way, the second experiment in Cutler et al. (1986) played the French wordlist used in Mehler et al. (1981) to a group of English-speaking listeners. (Note that syllable-monitoring is not a lexical task; therefore, in such a task, participants do not need to speak the language from which the lexical items have been selected. In fact, one may run syllabic-monitoring tasks using nonwords, as Cutler et al. (1986) did in their third experiment.) The task was the same as in the first experiment. The findings, once again, led to a null result—there was no target-by-word interaction. English listeners did not detect /bal/ faster in French *balcon* ‘balcony’ than in *ballon* ‘ball,’ neither they detected /ba/ faster in *ballon* than in *balcon*. Importantly, they did not detect /bal/ faster than /a/ in *balcon*. It seems that the difference between the French (Mehler, Dommergues, et al., 1981) and the English (Cutler et al., 1986) results were based on the participants rather than the materials.

The fourth (and final) experiment returned to French-speaking listeners. Cutler et al. (1986) asked whether French participants would demonstrate the use of syllabification in the segmentation of English materials, as it had been demonstrated with French materials. In order to run this experiment, the authors took the English auditory stimuli they had used in their first experiment and they played them to a group of French participants. Interestingly, this experiment led to a target-by-word interaction—French participants detected /ba/ faster in English *balance* than in *balcony*, and they detected /bal/ faster in *balcony* than in *balance*. In sum, it would appear that French-speaking listeners (but, crucially, not English-speaking listeners) exploit syllabic information in their segmentation of the speech signal, somewhat independently from the language of the materials.

At this juncture, two questions arise. First, are there other languages like French, in which syllabification is consistent and simple, whose speakers exploit syllabic segmentation during processing? Second, what is known about French-English bilinguals? For reasons of space, this chapter addresses exclusively the first question. Readers interested in bilingualism are referred to two articles, one on French-English bilinguals (Cutler, Mehler, Norris, & Seguí, 1992) and one on Spanish-English bilinguals (Bradley, Sánchez-Casas, & García-Albea, 1993, experiment 5). The articles discussed in the remainder of this section report on experiments designed to replicate the syllable-monitoring task of Mehler et al. (1981) with listeners who speak other Romance languages (including Spanish), all of which have syllabification patterns similar to those of French but also have some potentially important prosodic differences.

Sebastián-Gallés et al. (1992) observed that all the words used in the English materials of Cutler et al. (1986) were stress-initial, and all those used in the French materials of Mehler et al. (1981) were stress-final. This points to a confound between lexical stress and syllabicity—the

confound is due to lexical limitations of the languages involved rather than a design flaw. Catalan and Spanish, like English (but unlike French), have variable lexical stress—that is, unlike in French, stress is assigned lexically in these languages and does not consistently fall on the same syllable. For instance, /ba/ is stressed in *bala* ‘bullet’ but unstressed in *balada* ‘ballad,’ both of them Spanish words. Neither Catalan nor Spanish, like French (but unlike English), have ambisyllabicity. This means that, by exploring Catalan and Spanish, one may examine any potential interactions between stress and syllabic structure, which neither French nor English allow. Finally, Catalan, unlike Spanish, has unstressed vowel reduction. The experiments presented in Sebastián-Gallés et al. (1992) explored the possibility that lexical stress and unstressed vowel reduction play a role in the segmentation of the speech signal, perhaps interacting with syllabification.

The first experiment in Sebastián-Gallés et al. (1992) was run on Catalan. While Catalan has unstressed vowel reduction, the authors focused on vowel phonemes that do not reduce in unstressed position, /i/ and /u/, to examine suprasegmental, rather than segmental, stress effects. Importantly, the words they used could begin in either a stressed or an unstressed syllable. Otherwise, the design mimicked the one in Mehler et al. (1981). A group of Catalan participants were asked to monitor for sequences such as /pu/ or /pur/ in word-lists that included words such as *pura* ‘holy’ and *purga* ‘laxative’ (stressed on the first syllable), and *puré* ‘purée’ and *purgant* ‘depletory’ (stressed on the last syllable). The results yielded a target-by-word interaction, but only for words whose first syllable was unstressed—there was a null effect in the case of the words whose first syllable was stressed. Unstressed syllables are shorter, and, relative to those in stressed syllables, their segments are hyporarticulated (De Jong, 1995; Nadeu, 2014). According to Sebastián-Gallés and colleagues, listeners might need to rely more on top-down segmentation strategies—such as syllabification—when dealing with unstressed syllables, where the acoustic-segmental information is less reliable. This, in itself, could explain the difference between the English (Cutler et al., 1986) and the French (Mehler, Dommergues, et al., 1981) results discussed above; but, again, Catalan has unstressed vowel reduction, and the pattern obtained in this experiment might result from a strategy specific to the processing of languages with unstressed vowel reduction, even in full vowels.

In the case of Spanish, it will be seen that, while some experiments have yielded positive results, others have led to null results. There are two published articles on syllable-monitoring in Spanish (Bradley et al., 1993; Sebastián-Gallés, Dupoux, Seguí, & Mehler, 1992). In their second experiment, Sebastián-Gallés et al. (1992) manipulated, like in their Catalan experiment, the presence (vs. absence) of lexical stress in the first syllable of words used as auditory stimuli. Spanish-speaking participants from Barcelona were asked to monitor for sequences such as /ba/ and /bal/ in words such as *balón* ‘ball’ and *balcón* ‘raft’ (whose first syllable is unstressed), and *bala* ‘bullet’ and *balsa* ‘raft’ (whose first syllable is stressed). There were no positive effects—there was no target-by-word interaction. This was true for both stressed- and unstressed-initial words. The authors point out, however, that a comparable experiment with Spanish speakers and Spanish materials had found a syllabic effect, a target-by-word interaction (Sánchez-Casas, 1988). And, indeed, the second article, discussed below, did yield a positive result (Bradley et al., 1993).

Bradley and colleagues asked a group of Spanish-speaking participants from Madrid to monitor for sequences such as /pa/ and /pal/ in words such as *paloma* ‘dove’ and *palmera* ‘palm tree.’ This study did find a target-by-word interaction—/pa/ was detected faster in *paloma* than

in *palmera*, and /pal/ was detected faster in *palmera* than in *paloma*. This replicated the French findings and the “unstressed” Catalan findings, but was in contrast with the Spanish null finding reported in Sebastián-Gallés et al. (1992). In a follow-up experiment, however, Spanish-speaking listeners monitored English materials and, this time, they did not demonstrate the target-by-word interaction they had demonstrated when monitoring comparable Spanish materials. The latter finding contrasts with that in Cutler et al. (1986), which showed that French participants used syllabification when segmenting English materials. The findings lead us to ask the following: What explains the fact that Spanish listeners use syllabification strategies, but only for native materials (unlike French listeners, who used them for both native and foreign materials)? And, most importantly, why did Bradley et al. (1993) find an effect where Sebastián-Gallés et al. (1992) had found none? It would appear that syllabic effects, as a segmentation strategy, are especially vulnerable in Spanish—sometimes they are found and sometimes they are not.

One observation in place is that the average response times in Sebastián-Gallés et al. (1992) were between 350 and 400 milliseconds, while those reported in Bradley et al. (1993) were between 550 and 600 milliseconds. Recall that, while the former found no target-by-word interaction, the latter did. The tasks used in the two studies were identical, but there was a difference in the fillers (the words for which no response was expected because they did not include any of the target sequences). In Bradley et al. (1993), but not in Sebastián-Gallés et al. (1992), the fillers included words that were phonologically similar to the experimental words. This means that one task was cognitively more demanding than the other, and participants may have been more “careful” in one task than in the other. This explains the latency difference. But what is important is that the processing stages these two studies tapped into might have been different, leading to different results. Thus, one possibility is that, in Spanish, syllabic segmentation exists but takes place only in later, higher processing levels. When a task is completed at a very shallow processing level, syllabic effects might not come into effect. Indeed, in a final experiment, Sebastián-Gallés et al. (1992) replicated their own first experiment, but this time they told their participants that, at the end of the task, they would be given a recognition test including some of the words presented in the experiment. This slowed participants down. Interestingly, this time, syllabic effects emerged—there was an interaction between target type and word type.

Sebastián-Gallés and colleagues noted that, relative to French and Catalan, Spanish has fewer vowels, and it does not have unstressed vowel reduction. For native listeners, making phonological decisions (i.e., segmental categorization) about Spanish auditory stimuli may be easier than making them about French or Catalan materials, especially in the case of unstressed vowels—there are less candidate phonemes to choose from and less allophonic variability to filter out. In their words, Spanish is more “acoustically transparent” than French or Catalan. Perhaps this explains why, in Spanish-speaking listeners, syllabification is particularly vulnerable—i.e., it is part of the processing system, but it may be bypassed relatively easily when the acoustic evidence is clear enough to make an early decision. Note that, interestingly, syllabic segmentation also seems to be vulnerable among Italian listeners (Tabossi, Collina, Mazzetti, & Zoppello, 2000). In Italian, as in Spanish, some tasks yield it but others do not.

There seems to be no doubt that syllabification is part of Spanish speakers’ phonological knowledge; and this knowledge is exploited, at least under some processing conditions, during speech comprehension. Relative to French and Catalan, syllabic segmentation seems to be less robust in Spanish. Nonetheless, there is behavioral evidence suggesting syllabic segmentation in

all the Romance languages that have been examined to date: French, Catalan, Italian, and Spanish. It can be argued that speakers of Spanish and other Romance languages, but not English, “spontaneously tend to represent and segment speech signals in terms of syllables” (Sebastián-Gallés et al., 1992, p. 29).

Syllabic activation: word recognition and priming

The early “syllabic hypothesis” of Mehler and colleagues (Mehler, 1981; Mehler, Seguí, & Frauenfelder, 1981) holds that syllables are natural units in speech processing. Among other things, syllabic structure is hypothesized to modulate lexical access—i.e., initial contact with the mental lexicon is sifted by syllabic structure. But the evidence discussed so far comes from segmentation strategies, which are the ones revealed in syllable-monitoring and word-spotting tasks. While the results of segmentation-based experiments tend to support the hypothesis that (at least in some languages) syllabic structures affect speech processing, these studies “give no indication as to whether during the early phases of word recognition the first contact with the mental lexicon is actually established by a word’s initial syllable” (Tagliapietra, Fanari, Collina, & Tabossi, 2009, p. 513). Evidence from speech segmentation is informative, and it suggests that, in languages such as Spanish, syllables are indeed processing units and thus are part of the phonological knowledge of Spanish speakers. Nonetheless, it is theoretically relevant to know whether, beyond segmentation, syllables are employed in word activation. If they are, this could be interpreted to suggest that the phonological representation of words in the mental lexicon contains syllabic information, at least in some form.

Several experimental studies have addressed this issue, and evidence comes from at least two languages, Italian and Spanish. Two published articles deal with *spoken* word recognition data and they are both on Italian (Tabossi et al., 2000; Tagliapietra et al., 2009, p. 513), and a relatively large body of literature deals with *visual* word recognition data and focuses on Spanish (e.g., Álvarez, Carreiras, & Perea, 2004; Álvarez, Carreiras, & Taft, 2001; Álvarez, de Vega, & Carreiras, 1998; Carreiras, Álvarez, & de Vega, 1993; Carreiras & Perea, 2002).

A series of studies, of which two are reviewed in some depth, investigate the potential role of the syllable in Spanish visual lexical access—see Ferrand, Seguí, & Grainger (1996) for an example from French. The main conclusion of this body of literature is that words are indeed accessed syllabically. The majority of these studies typically contend that, in polysyllabic words, the first syllable is more prominent than the others in the process of activating lexical entries—the first syllable, for instance, determines to a certain extent the candidate words that enter the competition for lexical selection and the words that do not (e.g., Álvarez, Carreiras, & de Vega, 2000; Álvarez et al., 1998). It is not the case that reading proceeds strictly from left to right; the evidence, however, does suggest that word-initial letters play a privileged role in visual word recognition (Perea, 1998). To the extent that phonology is involved in visual lexical access (e.g., Pollatsek, Perea, & Carreiras, 2005), findings from visual word recognition may help us understand phonological processing and, in turn, phonological representations.

In studies that maintain that lexical activation processes privilege the first syllable, it is usually found that words beginning with a high-frequency syllable—that is, those whose first syllable is shared with many other words—take longer to be recognized than words beginning with low-frequency syllables—that is, those whose first syllable is shared with few other words (Carreiras et al., 1993; Carreiras & Perea, 2002). The explanation is that the first syllable

activates the entire cohort of words that begin with it, but not those that do not. In large cohorts, lexical competition amongst the activated candidates triggers the recognition delay: as the size of the cohort increases, so does the competition. However, rather than the sheer number of syllabic neighbors, follow-up studies have found that what modulates a word's recognition latency is the token frequency of its neighbors. In other words, a word with higher-frequency syllabic neighbors—that is, syllabic neighbors with a higher token frequency than the target word itself—will take longer to be recognized than a word with lower-frequency syllabic neighbors (Carreiras & Perea, 2002; Perea & Carreiras, 1995, 1998). On the basis of the available evidence, these studies tend to conclude, along with Álvarez, Carreiras, & Taft (2001), that “any model of lexical access has to incorporate a syllabic model of representation or include the syllable as a sublexical unit of processing in Spanish” (page 553).

Carreiras & Perea (2002) reported on four visual lexical decision tasks with Spanish materials. In visual lexical decision tasks, participants are shown words and nonwords (in their language) in random order, and they are asked to press a button if they recognize an actual word in the string of letters they see in a particular trial. The four lexical decision tasks in Carreiras & Perea (2002) used the masked priming paradigm. In this paradigm, a trial comprises a forward mask (four pound signs displayed for 500 milliseconds, in this case), a prime (a word or nonword displayed in lowercase), which, crucially, is shown for a very short period of time, and a target (usually a word or nonword in capital letters). A prime is a word form acting as a cue or intervention preceding a target; the target is the word form where the measurement is taken or observation made. Participants, who are not warned of the presence of the mask or the prime, perform lexical decisions only on the targets, and it is typically reported that they are unaware of the presence of the primes. With this paradigm, researchers investigate how some experimentally-manipulated, structural aspect of the prime affects (or does not affect) the target's recognition latency. Since participants are typically unaware of the presence of the primes due to the fact that they are shown just for a few milliseconds, masked priming experiments tap into unconscious lexical and prelexical processes, including phonological encoding.

The first experiment in Carreiras & Perea (2002) had two conditions: A target such as *bono* ‘bond’ was preceded by either an unrelated control prime, such as *caja* ‘box’ (with which it does not share the initial syllable), or an experimental prime, such as *boca* ‘mouth’ (with which it shares the first syllable). In this case, the experimental prime is a Spanish word of a higher token frequency than the target. Relative to unrelated controls, orthographically similar primes tend to facilitate recognition, and thus accelerate response latencies. In this experiment, however, the authors found that, relative to unrelated controls (*caja-BONO*), their experimental primes (*boca-BONO*) delayed the recognition of target words. Arguably, the related prime (*boca*) and the target (*bono*) compete for recognition—they are simultaneously activated because they are syllabic neighbors. In this experiment, since primes were chosen to have higher token frequencies than targets, recognition rates were delayed rather than accelerated—this is known as inhibition.

Nonwords have the potential to facilitate the recognition of words if they share crucial phonological material, such as the first syllable, but they cannot compete for recognition, as they are not stored in the lexicon—they cannot, by definition, trigger inhibition. In their second experiment, Carreiras & Perea (2002) used nonword primes in two experimental conditions: one in which a target, such as *bono* ‘bond,’ was preceded by a nonword with which it shares the first syllable (*bopa*) and one in which it does not (*caya*). It was found that the recognition latencies of

the targets were shorter in the experimental (*bopa*) than in the control (*caya*) condition. Evidently, words are recognized faster after having seen nonwords with which they share the first syllable, which suggests that activating a syllabic item (*bo-*) facilitates the access of words that begin with that syllable. But the inhibition and facilitation effects found in these experiments may have an easier explanation—the effects may be due, not to the first syllable *per se*, but to the first two letters. Two additional experiments address this issue.

First, the authors compared the facilitation of the *bopa-BONO* condition (relative to the control, *caya-BONO*) with two additional conditions, one in which a monosyllabic four-letter word (e.g., *zinc* ‘zinc’) was preceded by a monosyllabic four-letter nonword (*ziel*) with which it shares the first two letters but, crucially, not the entire syllable (*ziel-ZINC*), and one in which the monosyllabic four-letter word is preceded by an unrelated nonword (*flur*). This experiment did not find any facilitation effects for the *ziel-ZINC* condition. It seems, therefore, that a word’s cohort is determined by the first syllable, and not merely the first two segments. However, as the authors point out, monosyllabic content words are unusual in Spanish, and processing differences between disyllabic and monosyllabic words must be interpreted with care.

The last experiment in Carreiras & Perea (2002) utilized the partial masked priming paradigm and focused on the difference between closed and open syllables. The targets were six-letter words with either a closed (*bastón* ‘cane’) or an open (*basura*) word-initial syllable; importantly, the first three letters were identical in both cases, but the third letter was either syllabified with the first or with the second syllable. Partial masked primes were either two-letter (*ba####*) or three-letter (*bas####*) sequences followed by pound signs. If lexical activation takes into account syllabic configuration, one would expect that two-letter partial primes facilitate words such as *basura* while three-letter partial primes facilitate words such as *bastón*. Indeed, lexical decision times to *bastón* were shorter after *bas*-type than after *ba*-type partial primes, and recognition latencies to *basura* were shorter after *ba*-type than after *bas*-type partial primes. By hypothesis, when participants see *ba#####*, they create a syllabic “mental image” consisting of a closed syllable, which facilitates the activation of words beginning with that particular closed syllable. This experiment suggests that *bastón* and *bastante* ‘enough,’ but not *basura*, are part of the same syllabic cohort, which, in turn, suggests that syllabic structure is (somehow) part of the representation of words in the mental lexicon.

Even though the evidence comes from reading experiments, “syllabic effects on visual word recognition have usually been interpreted as involving phonological processing” (Álvarez et al., 2004, p. 428). But this needs to be tested directly, rather than assumed. In order to explore whether phonological processing is at the root of the syllabic effects found in previous experiments, Álvarez, Carreiras, & Perea (2004) ran three visual word recognition experiments similar to the ones reported in Carreiras & Perea (2002), but with a crucial twist. The main idea in this new set of studies stems from the fact that, although the Spanish spelling system is shallow, it is not transparent. The first experiment is a replication of prior work with a similar, but not identical, experimental paradigm. This experiment tested whether the activation of five-letter words, such as *junio* ‘June,’ may be facilitated by the presentation of masked primes (nonwords) with which they share the first three letters but not the first syllable—*junio* was preceded by either *juntu* or *junas*. It was found that words such as *junio* were indeed facilitated by nonwords such as *junas*, with which they share the first syllable, but not by words such as *juntu*, with which they do not.

Álvarez et al.'s (2004) second experiment added a crucial condition. The conditions in the first experiment exploited the orthographic and phonological similarities between nonword primes and target words. The condition added in the second experiment broke the orthographic similarity but retained the phonological one—the target *girar* ‘to turn’ could be preceded, not only by *girol* or *girten*, but also by *jirol* or *jirten*. Note that, in Spanish, both <gi> and <ji> correspond to /xi/. The first pair of primes share with the target word not only the pronunciation of the first three letters, but also the letters themselves; the second pair of primes, on the other hand, share with the target word only the pronunciation of the first three letters, but not the letters themselves. The results showed that both conditions triggered the expected syllabic effects. Thus, upon seeing a nonword such as *jirol*, readers create a “phonological image” of this nonword (i.e., reading the word activates the phonological system), and it is this phonological image that facilitates the recognition of words with which the nonword shares the first phonemes (and, crucially, the first syllable). In sum, a word’s cohort, activated during visual and spoken word recognition, comprises its syllabic neighbors, the words with which it shares the first phonological (not orthographic) syllable.

We are not aware of any Spanish *spoken* word recognition experiments focusing on the syllable, but two studies on Italian seem to be relevant (Tabossi et al., 2000; Tagliapietra et al., 2009, p. 513). Tabossi et al. (2000) reported on five experiments, four of which dealt with syllable- and phoneme-monitoring data; the fifth experiment was concerned with lexical activation and the potential role of the syllable, and that is the one discussed here. This experiment utilized the lexical decision task in a cross-modal priming paradigm. In this paradigm, target words (the ones on which participants conduct lexical decisions) are presented visually, but primes are presented auditorily. In such tasks, participants are instructed to focus only on the visual targets, but they cannot avoid hearing the primes. In Tabossi et al.’s (2000) fifth experiment, primes were partial words. The partial primes consisted of three segments, always a consonant + vowel + consonant sequence, that would have been extracted from two types of words: (i) words whose second consonant is part of the first syllable (closed condition, such as *silvestre* ‘silvan, of the woods’), and (ii) words whose second consonant is part of the second syllable (open condition, such as *silenzio* ‘silence’). Targets were selected from among the semantic associates of the words from which primes had been extracted—e.g., *silenzio* ‘silence’ and *rumore* ‘noise’ are associates. For instance, participants would hear *La prossima parola è sil-* ‘the next word is sil-’ in two conditions (one [sil]-sequence extracted from *silvestre* and one from *silenzio*), and immediately see RUMORE; they were to make a lexical decision on RUMORE. If [sil] facilitated the activation of *silenzio*, RUMORE would be primed relative to a control condition. This experiment found, indeed, that [sil] from *silenzio* (but, crucially, not from *silvestre*) primed RUMORE, and [sol] from *soldato* ‘soldier’ (but, crucially, not from *solare* ‘solar’) primed GUERRA ‘war.’ Tabossi et al. (2000) tested only words whose first syllable is unstressed, but Tagliapietra et al. (2009) extended these findings to words whose first syllable is stressed. The authors conclude that syllabic structure is involved in lexical activation, and it would follow that it is part of the mental lexicon: “when the syllabic structure of a sequence of phonemes is the same as the word initial structure of a word that is semantically related to the target, the sequence effectively primes the visual target” (Tabossi et al., 2000, p. 770). Thus, [sil] from *silvestre* and [sil] from *silenzio* are phonetically different, and this difference is consistent enough to have formed a pattern. Listeners are finely tuned to these differences, which stem from syllabification (among other things), and they exploit them in their lexical searches.

Tangentially, these findings also suggest that mental representations of word forms are not comprised merely of abstract phonemes (as typically assumed in formal phonological models), but of segments with detailed phonetic content.

The findings in the studies discussed so far in this section are strongly in line with the hypothesis that syllabification is exploited in lexical access, and syllabic configurations are included in a word's mental representation. The last study to be discussed, on the other hand, provides some counter evidence to this hypothesis. In Tagliapietra et al.'s (2009) second experiment, Italian speakers participated in a lexical decision task with the cross-modal priming paradigm. Auditory primes were two-segment sequences, rather than three-segment ones. The setup was very similar to Tabossi et al.'s (2000) fifth experiment. The authors reasoned that, if a word's cohort is formed only by its syllabic neighbors, a two-segment sequence such as [ku] could not prime *curva* 'bend,' but only *cura* 'treatment.' (It had already been demonstrated that [sol] primed *soldato* 'soldier,' but not *solare* 'solar.')

In this experiment, two-segment sequences extracted out of words whose first syllable is closed—e.g., [ku] cut out from *curva*—were found to prime semantic associates of words whose first syllable is open. In other words, a sequence such as [ku] extracted from *curva* was found to facilitate the word MEDICO 'doctor,' which is an associate of *cura* 'treatment' but not *curva* 'bend.' This study demonstrates that a word's cohort is not restricted to the words with which it shares an entire word-initial syllable. Lexical access may exploit sub-syllabic information. These results may be in contrast with those in Carreiras and Perea's (2002) last experiment, but note the differences in the experimental setup and the language tested. The evidence presented in Tabossi et al. (2000) and Tagliapietra et al. (2009) suggest that Italian speakers may use syllabic information during word recognition, but they are not restricted to it, as they may exploit other information, including sub-syllabic information.

The studies reviewed in this section have been concerned with the potential role of the syllable in the activation and recognition of visual words. These studies have all used the lexical decision task with either cross-modal (auditory) or monomodal (visual) priming, and they have investigated two very similar languages, Spanish and Italian. It has been found that, in the process of recognizing words, the syllabic similarities of primes and target words modulate (but do not fully restrict) lexical activation. In conditions in which primes and targets present matching syllabic configurations, primes facilitate the recognition of lexical targets by reducing response times. It follows that syllabic structures (syllabification) are important in the process of recognizing words, but parallel processes are available. It also follows that the phonological representation of words, in Spanish and Italian, must include syllabic information. The syllable is an important aspect of the phonological knowledge of speakers of Spanish and Italian, and it would seem that the entries in the lexicon include such syllabic information.

CONCLUSION

The present chapter has reviewed some of the evidence suggesting that syllabification is part of the phonological knowledge of Spanish speakers as well as that of speakers of other Romance languages, such as French, Catalan, and Italian. Some of the experimental findings discussed here suggest that listeners who speak Spanish exploit their knowledge of their language's syllable structure to dynamically segment the speech signal into chunks that may then be used to initiate lexical searches. This is not their only segmentation strategy, but it is one of them. In

other words, speakers of Romance languages such as French and Spanish are not necessarily restricted to syllable-based segmentation, but this segmentation process is part of their repertoire. Other experimental findings suggest that the strategy Spanish listeners use to select (or activate) lexical items, during word recognition, is (at least partially) based on the syllabification patterns of this language. Again, the lexicon may be accessed in more than one way, but syllable-based activation is one of them. That is, speakers of Romance languages such as Italian and Spanish are not restricted to syllable-based lexical access, but this lexical-activation process is part of their repertoire. Finally, some evidence also suggests that long-term phonological representations of lexical items (also known as *phonolexical* forms or underlying representations) include information about their syllable structure. The nature of this knowledge has yet to be determined.

Importantly, it appears that not all languages exploit syllabification as a segmentation or word-activation strategy, and it is not clear that syllabic configuration is part of the long-term representation of words in all languages. A case in point is that of English, for which experimental evidence has failed to find any performance-based evidence of syllabic segmentation and syllable-based lexical access. This is not to say that syllables are not part of the phonology of English, but it appears at least that the complexity of English syllabic patterns encourage English listeners to ignore syllabification during the online segmentation of the speech signal, perhaps because it might not be as efficient a strategy as it is in other languages. This is reminiscent of the facts regarding lexical stress. While lexical stress is contrastive in both Spanish and English (that is, it is used to distinguish words, as in English *trusty* and *trustee*, and in Spanish *bebe* ‘s/he drinks’ and *bebé* ‘baby’), only Spanish listeners exploit lexical stress during spoken word recognition (Soto-Faraco et al., 2001). This might be so because English lexical stress is mostly word-initial (and therefore mostly predictable) or because English lexical stress tends to be cued by vowel timbre and only sporadically it is not (Cutler, 2005; van Donselaar, Koster, & Cutler, 2005).

The most important contribution of this chapter is that it illustrates that behavioral evidence extracted from speech comprehension tasks, such as segmentation- and activation-based paradigms, may provide important information to phonologists. In addition to examining the phonetics of speech production, laboratory phonologists must investigate the speech comprehension process. During speech comprehension, listeners engage in a process known as phonological encoding, which consists of processes such as detecting word onsets (segmentation), assigning acoustic information to phonemic categories (categorization), and searching words in the lexicon (lexical activation). Evidence gathered from tasks that tap into phonological encoding may prove to be relevant for our understanding of speakers’ mental representations, which is the object of phonology. In writing this chapter, we decided to illustrate the importance of comprehension-based phonological evidence by focusing on a single issue, the role of the syllable. The chapter has argued that the evidence to date suggests that Spanish speakers engage in syllable-based segmentation strategies. This demonstrates that they possess some type of mental representation of what is a possible (and impossible) Spanish syllable. This knowledge is deployed when segmenting the speech signal (including words and nonwords) into parsable chunks. Spanish speakers also engage in syllabification-based word activation—that is, syllabic knowledge is part of the mechanism that helps find words in the mental lexicon. The present review concludes by posing the following question: Are there other structures of the phonology of Spanish that may receive support (or, on the contrary, may be falsified) by the phonological-encoding evidence?

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